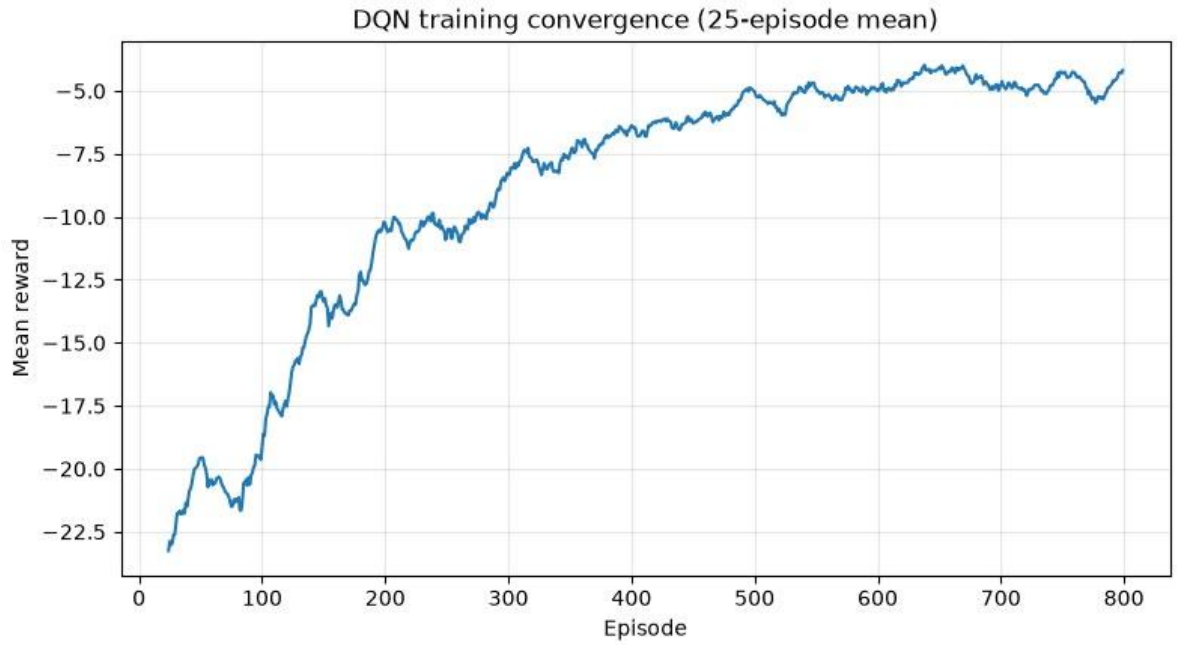
DQN Training Convergence



Graph 1: DQN training convergence. Needs artifacts/training_rewards

```
import argparse
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np

parser = argparse.ArgumentParser()
parser.add_argument("--artifacts", default="artifacts")
args = parser.parse_args()
out = Path(args.artifacts)

rewards = np.loadtxt(out / "training_rewards.csv", delimiter=",", skiprows=1)[:, 1]
window = min(25, len(rewards))
moving = np.convolve(rewards, np.ones(window) / window, mode="valid")
```

```
fig, ax = plt.subplots(figsize=(8, 4.5))
ax.plot(np.arange(len(moving)) + window - 1, moving, color="tab:blue")
ax.set(title="DQN training convergence (25-episode mean)", xlabel="Episode", ylabel="Mean reward")
ax.grid(alpha=0.3)
fig.tight_layout()
fig.savefig(out / "plot1_training_convergence.png", dpi=180)
plt.close(fig)
print("saved", out / "plot1_training_convergence.png")
```